KAIZEN MATHS

Lesson starter

Linear Algebra: Systems, Span, Basis and Transformations

Systems and Row Operations

Name

S1

Solve each system using elimination or row operations.

Solve the system −4x − 5y = 8, −3x − 3y = 3.

S2

Carry out the stated row operation on the augmented matrix.

Carry out $R_2 \leftarrow R_2 + 2R_1$ on $\left[\begin{array}{cc|c} 1 & 3 & 3 \\ 0 & 2 & 4 \end{array}\right]$.

S3

Write each system in augmented matrix form.

Write the augmented matrix for 2x − y − 4z = 4, −x + y − z = 9, −x − 5y − 3z = 6.

S4

Solve each triangular system by back substitution.

Solve the triangular system 4x + 2y + z = 4, 3y − z = 5, 3z = −6.

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Systems and Row Operations

Name _____

EA1

Solve each system using elimination or row operations.

Solve the system −5x − 2y = −35, 5x − y = 20.

EA2

Carry out the stated row operation on the augmented matrix.

Carry out $R_2 \leftarrow R_2 - 3R_1$ on $\left[\begin{array}{cc|c} -4 & -4 & -2 \\ -3 & 3 & -6 \end{array}\right]$.

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Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Systems and Row Operations

Name _____

EB1

Write each system in augmented matrix form.

Write the augmented matrix for 2x + 5y − 2z = 1, 3x + 4y − 3z = 3, −5x − 2y + 5z = −7.

KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Systems and Row Operations

Name _____

EA1

Solve each system using elimination or row operations.

Solve the system −5x − 2y = −35, 5x − y = 20.

EA2

Carry out the stated row operation on the augmented matrix.

Carry out $R_2 \leftarrow R_2 - 3R_1$ on $\left[\begin{array}{cc|c} -4 & -4 & -2 \\ -3 & 3 & -6 \end{array}\right]$.

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KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Systems and Row Operations

Name _____

EB1

Write each system in augmented matrix form.

Write the augmented matrix for 2x + 5y − 2z = 1, 3x + 4y − 3z = 3, −5x − 2y + 5z = −7.

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Show your working

Teacher answers & working

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S1

Solve each system using elimination or row operations.

Solve the system −4x − 5y = 8, −3x − 3y = 3.

Answer:

$$\begin{bmatrix} 3 \\ -4 \end{bmatrix}$$

Working:

Write the augmented matrix for the system.

$$\left[\begin{array}{cc|c} -4 & -5 & 8 \\ -3 & -3 & 3 \end{array}\right]$$

Eliminate x using $R_2 \leftarrow -4R_2 + 3R_1$.

$$\left[\begin{array}{cc|c} -4 & -5 & 8 \\ 0 & -3 & 12 \end{array}\right]$$

$-3y = 12$, so $y = -4 = -4$.

Substitute y = −4 into −4x − 5y = 8.

$-4x + 20 = 8$, so $-4x = -12$ and $x = 3 = 3$.

Solution vector: $\begin{bmatrix} 3 \\ -4 \end{bmatrix}$.

S2

Carry out the stated row operation on the augmented matrix.

Carry out $R_2 \leftarrow R_2 + 2R_1$ on $\left[\begin{array}{cc|c} 1 & 3 & 3 \\ 0 & 2 & 4 \end{array}\right]$.

Answer:

$$\left[\begin{array}{cc|c} 1 & 3 & 3 \\ 2 & 8 & 10 \end{array}\right]$$

Working:

Keep R_1 unchanged.

New R_2 first entry: 0 + 2(1) = 2.

New R_2 second entry: 2 + 2(3) = 8.

New R_2 constant: 4 + 2(3) = 10.

$$\left[\begin{array}{cc|c} 1 & 3 & 3 \\ 2 & 8 & 10 \end{array}\right]$$

Teacher answers & working

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S3

Write each system in augmented matrix form.

Write the augmented matrix for 2x − y − 4z = 4, −x + y − z = 9, −x − 5y − 3z = 6.

Answer:

$$\left[\begin{array}{ccc|c} 2 & -1 & -4 & 4 \\ -1 & 1 & -1 & 9 \\ -1 & -5 & -3 & 6 \end{array}\right]$$

Working:

The coefficient matrix uses the coefficients of x, y, and z in order.

$$A = \begin{bmatrix} 2 & -1 & -4 \\ -1 & 1 & -1 \\ -1 & -5 & -3 \end{bmatrix}$$

The constants form the final augmented column.

$$\left[\begin{array}{ccc|c} 2 & -1 & -4 & 4 \\ -1 & 1 & -1 & 9 \\ -1 & -5 & -3 & 6 \end{array}\right]$$

S4

Solve each triangular system by back substitution.

Solve the triangular system 4x + 2y + z = 4, 3y − z = 5, 3z = −6.

Answer:

$$\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

Working:

Start from the last equation because it contains only z.

$3z = -6$, so $z = -2 = -2$.

Substitute z = −2 into 3y − z = 5.

$3y + 2 = 5$, so $3y = 3$ and $y = 1 = 1$.

Substitute y = 1 and z = −2 into 4x + 2y + z = 4.

$4x + 2 - 2 = 4$, so $4x = 4$ and $x = 1 = 1$.

Solution vector: $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$.

Teacher answers & working

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EA1

Solve each system using elimination or row operations.

Solve the system −5x − 2y = −35, 5x − y = 20.

Answer:

$$\begin{bmatrix}5\\5\end{bmatrix}$$

Working:

Write the augmented matrix for the system.

$$\left[\begin{array}{cc|c}-5 & -2 & -35\\ 5 & -1 & 20\end{array}\right]$$

Eliminate x using $R_2 \leftarrow -5R_2 - 5R_1$.

$$\left[\begin{array}{cc|c}-5 & -2 & -35\\ 0 & 15 & 75\end{array}\right]$$

$15y = 75$, so $y = 5 = 5$.

Substitute y = 5 into −5x − 2y = −35.

$-5x - 10 = -35$, so $-5x = -25$ and $x = 5 = 5$.

Solution vector: $\begin{bmatrix}5\\5\end{bmatrix}$.

EA2

Carry out the stated row operation on the augmented matrix.

Carry out $R_2 \leftarrow R_2 - 3R_1$ on $\left[\begin{array}{cc|c}-4 & -4 & -2\\ -3 & 3 & -6\end{array}\right]$.

Answer:

$$\left[\begin{array}{cc|c}-4 & -4 & -2\\ 9 & 15 & 0\end{array}\right]$$

Working:

Keep R_1 unchanged.

New R_2 first entry: −3 − 3(−4) = 9.

New R_2 second entry: 3 − 3(−4) = 15.

New R_2 constant: −6 − 3(−2) = 0.

$$\left[\begin{array}{cc|c}-4 & -4 & -2\\ 9 & 15 & 0\end{array}\right]$$

Teacher answers & working

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EB1

Write each system in augmented matrix form.

Write the augmented matrix for 2x + 5y − 2z = 1, 3x + 4y − 3z = 3, −5x − 2y + 5z = −7.

Answer:

$$\left[\begin{array}{ccc|c} 2 & 5 & -2 & 1 \\ 3 & 4 & -3 & 3 \\ -5 & -2 & 5 & -7 \end{array}\right]$$

Working:

The coefficient matrix uses the coefficients of x, y, and z in order.

$$A = \begin{bmatrix} 2 & 5 & -2 \\ 3 & 4 & -3 \\ -5 & -2 & 5 \end{bmatrix}$$

The constants form the final augmented column.

$$\left[\begin{array}{ccc|c} 2 & 5 & -2 & 1 \\ 3 & 4 & -3 & 3 \\ -5 & -2 & 5 & -7 \end{array}\right]$$